

Restaurant data analysis

Title: Restaurant data analysis

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Internship Program: Data Science Internship

Introduction

In today's digital world, restaurant platforms generate huge amounts of data related to customer preferences, ratings, pricing, and cuisine types. This project focuses on analyzing restaurant data using Exploratory Data Analysis (EDA) techniques to extract meaningful insights and business intelligence.

Objective

The main objective of this project is to analyze restaurant data and understand:

- Customer rating patterns
- Popular cuisines
- Restaurant distribution across locations
- Relationship between cost and rating
- Customer engagement using votes

Dataset description

The dataset contains 56,252 records with the following attributes:

- Restaurant name and address
- Location
- Cuisines offered

- Rating (rate)
- Votes (votes)
- Cost for two people (cost)
- Online order and table booking availability
- Restaurant type

Tools & Technology used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- WordCloud

Data Preprocessing & Cleaning

The dataset was cleaned using the following steps:

- Handled missing values in categorical columns using placeholders
- Converted rate, votes, and cost columns into numeric format
- Removed unwanted symbols such as “/5”, commas, and spaces
- Filled missing numeric values using median
- Dropped unnecessary columns such as phone numbers

These steps ensured the dataset was suitable for analysis.

Exploratory Data Analysis (EDA)

❖ Cuisine Analysis

- Cuisines were split and analyzed individually
- Certain cuisines showed higher average ratings
- Popular cuisines dominate customer preferences

❖ Location Analysis

- Restaurants are heavily concentrated in specific locations
- Some areas act as restaurant hotspots
- These locations show higher customer activity

❖ Price vs Rating Analysis

- Weak correlation observed between cost and rating
- Expensive restaurants are not always highly rated
- Medium-priced restaurants often perform better

❖ Votes vs Rating Analysis

- Higher votes generally indicate popular restaurants
- Votes and ratings together help identify trusted restaurants

Visualization techniques used

The following visualizations were created:

- Bar charts for cuisine and location analysis

- Scatter plots for cost vs rating and votes vs rating
- Heatmap for correlation between numerical features
- WordCloud for most popular cuisines

Key Insights

- Certain locations have a high concentration of restaurants
- Medium-cost restaurants tend to receive better ratings
- Cuisine type strongly influences customer satisfaction
- Votes are a strong indicator of restaurant popularity

Business Recommendations

- Focus on improving visibility in high-density locations
- Build cuisine-based recommendation systems
- Promote budget-friendly but highly rated restaurants
- Use combined ranking of votes and ratings for better accuracy
- Highlight popular dishes to increase engagement

Conclusion

This project successfully demonstrates how exploratory data analysis can transform raw restaurant data into meaningful insights. These insights can help businesses improve recommendation systems, enhance customer experience, and make data-driven decisions.

Top Cuisines by Rating



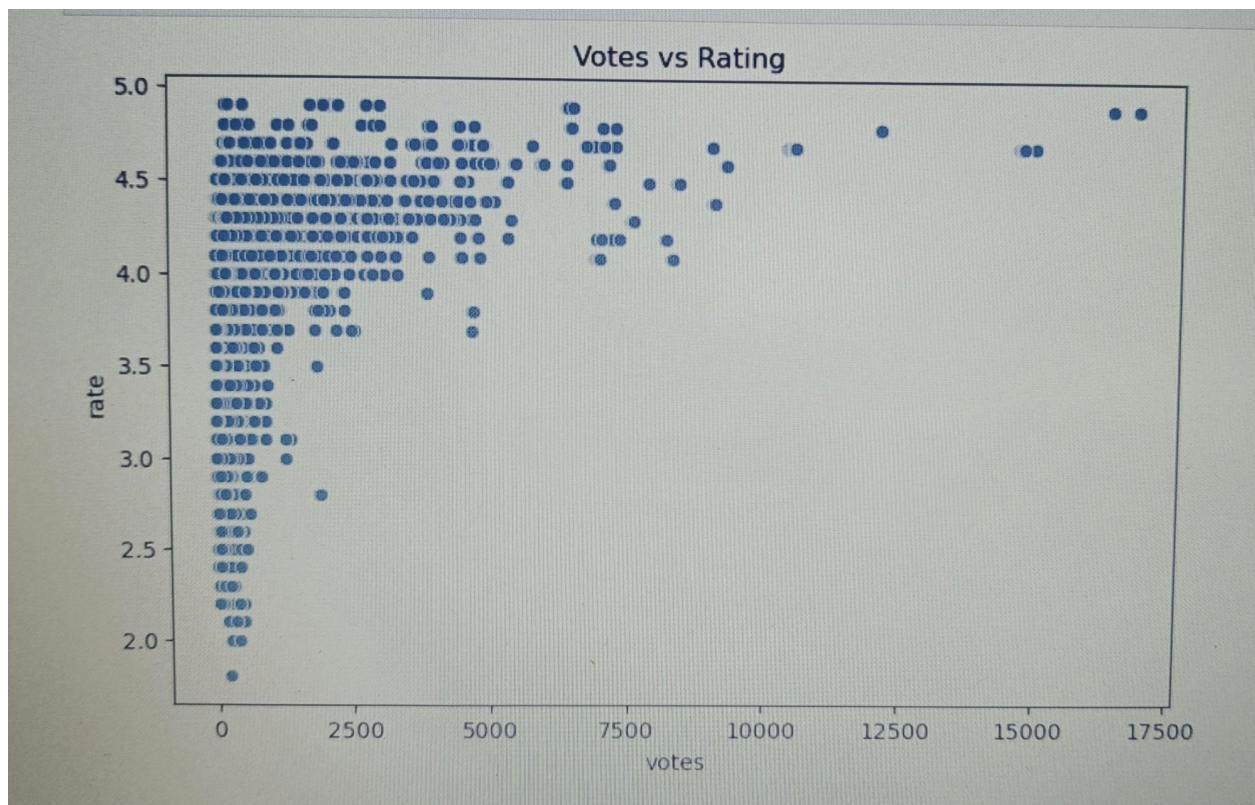
Top Restaurant Locations



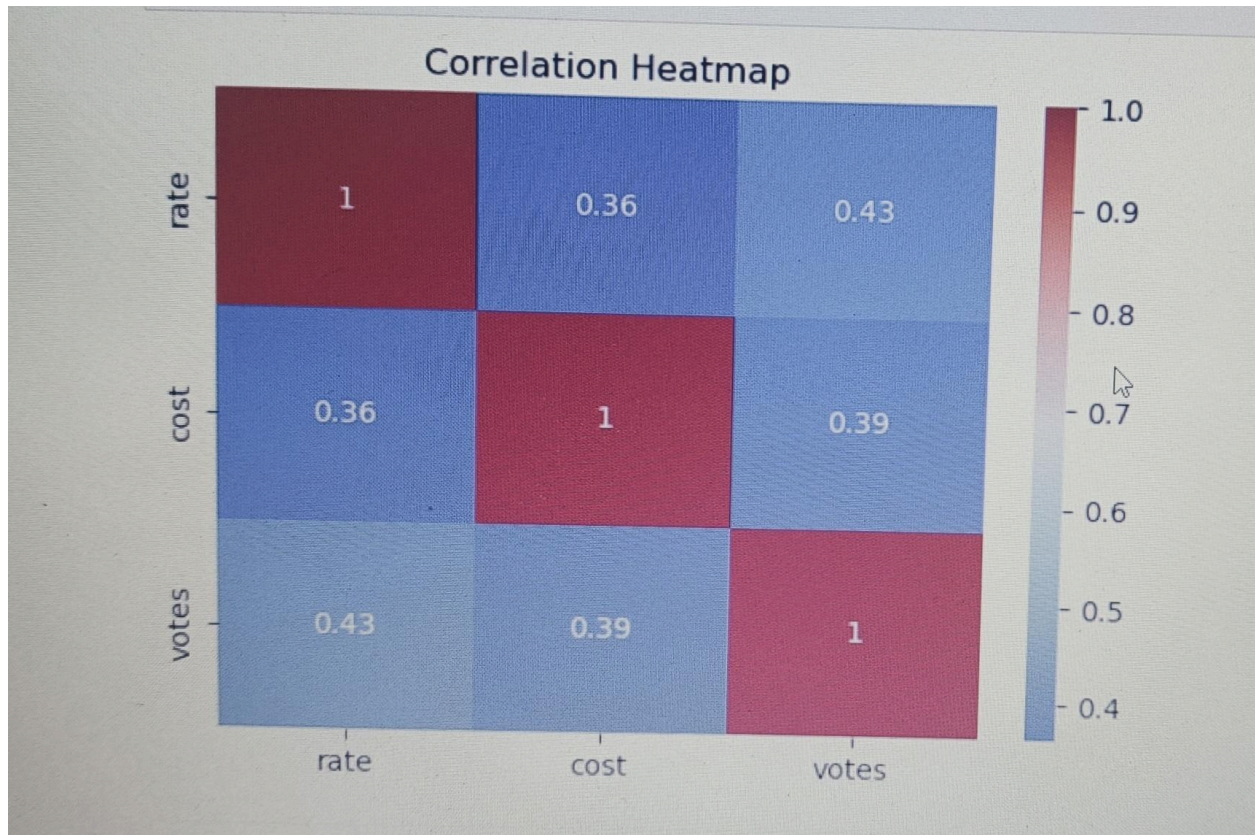
Cost vs Rating



Votes vs Rating



Correlation Heatmap



Popular Cuisines

[illegible]

Conclusion

```
print(Average cost. , df[ 'cost' ].mean())
```

Top rated restaurant name: Byg Brewski Brewing Company
Most common location: BTM
Most popular cuisine: North Indian
Average rating: 3.7003324326246183
Average cost: 541.9447308540141

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